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1. Scope

The Risk Management Report contains the outputs and summary of risk management activities. The general planning and methods are described in the Risk Management Plan, while the actual risks are listed and analysed in the Risk Management Matrix.

Product information	Description
Device family name	VR-CARDIO
Intended use	VR-CARDIO family is intended for use by electrophysiologists and cardiology specialists outside the operating room as an aid to real-time visualization and interpretation of three-dimensional electro anatomical maps of cardiac electrophysiology. It provides an immersive, three-dimensional stereoscopic view of the electrical behavior of the heart, enhancing non-invasive mapping capabilities during the evaluation of conditions such as atrial fibrillation and other arrhythmias. The system enables the analysis, visualization and informative interpretation of cardiac electrophysiological data, aiding in the understanding of complex arrhythmias and guiding preliminary assessments. It is beneficial in assisting and treating patients at risk of or being evaluated for cardiac arrhythmias. Its use is intended to assist traditional diagnostic methods and to deepen the understanding of arrhythmia mechanisms in order to develop non-invasive management strategies. The VR-CARDIO family consists of two main components: • VR-CARDIO Explore (Ref. VR-CARDIO-01) is a web page that receives the raw data, stores it, launches the algorithms to process the data and uploads it to the database so that the VR-CARDIO VHET360 can access it. • VR-CARDIO VHET360 (Ref. VR-CARDIO-02) is a software system designed for the real-time visualization and interpretation of three-dimensional electroanatomical maps of cardiac electrophysiology. It assists cardiology specialists in diagnosing and treating arrhythmias such as atrial fibrillation.
Market history	There is no market history. The VR-CARDIO is not yet sold in any territory.
Product code	VR-CARDIO-01 VR-CARDIO-02

2. Definitions / acronyms

Risk: combination of the probability of occurrence of harm and the severity of that harm.

Risk management: systematic application of management policies, procedures and practices to the tasks of analysing, evaluating, controlling and monitoring risk.

Injury: physical lesion, health damages, property or environmental damages.

Harm: Health damage, including the damage that can occur from loss of product quality or availability.

Hazard: The potential source of harm.

Hazardous Situation: circumstance in which people, property or the environment are exposed to one or more hazards (ISO 14971)

Adverse event: product defect (i.e. malfunction or failure, deterioration in characteristics or performance, or inadequacy of labelling or of instructions for use) that, directly or indirectly, has led or might have led to serious consequences, namely deterioration in the state of health of the patient, user or another person. Also called an incident.

Severity: A measure of the possible consequences of a hazard.

Probability of Occurance: Rank the probability of a failure occurring during the conception, manufacturing and commercialization of the product

Risk Assessment: Set of operations that allows the evaluation of the potential harm (damage) or injury.

Risk Analysis: Study that aims to determine the possible risks that may occur during the operational phase of a product.

Risk Control: Application of a set of operations that allows evaluation, control and monitoring of risks associated with a particular product or process in order to ensure that the identified risks do not occur or has a very low probability.

Residual Risk: Risk remaining after risk control measures have been taken.

Potential Hazard Identification and Assessment Team (PHIAT): Representatives of several functions that contribute to risk evaluation according to their specific knowledge.

RMT: Risk Management Team

RI: Risk Index

S: Severity

P: Probability of Occurrence

RMP: Risk Management Plan

RMR: Risk Management Report

3. References / bibliography

- Regulation (EU) 2017/745 on Medical Devices
- Standard EN ISO 13485:2016, point 7.1
- ISO 14971:2019 Medical devices – Application of risk management to medical devices
- ISO/TR 24971:2020 - Medical devices — Guidance on the application of ISO 14971
- IEC 62366:2015 - Medical devices – Application of usability engineering to medical devices.
- IEC/TR 80002-1:2009 Medical device software Part 1: Guidance on the application of ISO 14971 to medical device software
- IEC 80001-1:2021: Safety, effectiveness and security in the implementation and use of connected medical devices or connected health software – Part 1: Application of risk management
- Internal documentation: Quality Manual; Procedures; Clinical Evaluation report, Usability Report, VR-CARDIO Technical Documentation

4. Risk analysis

For each potential hazard a risk estimate can be calculated based on the following parameters:

Severity – S and **P:** Probability of Occurrence - P

See detailed information in Annex I (Risk Analysis) and Annex II (SOUP list).

5. Risk control measures

Risks were reduced as far as possible (AFAP). If a risk was classified as “Tolerable” or “Unacceptable” based on the Risk Analysis, Risk Control Measures were implemented. Potential categories of Risk Control Measures are “inherent safety by design”, “protective measures” and “information for safety”. In total, 124 risk control measures (112 on risk analysis and 12 on SOUP list) were implemented. These measures include reference to the instruction of use, user training, product release conditions control, test plans, etc.

See detailed information in Annex I (Risk Analysis) and Annex II (SOUP list).

6. Risk acceptability

The risk is estimated for each potential harm, resulting in the combination of the following factors: Severity (S) and Probability of Occurrence (P) multiplying the different weights for those factors to obtain the RPN (Risk Priority Number).

$$RPN = P \times S$$

Each risk estimated must be systematically evaluated, considering that risks should be reduced as far as possible without adversely affecting the benefit-risk ratio.

Therefore, even if the risk is acceptable, it is necessary to implement measures to bring the risk to its lowest level. Note that is not only enough to bring the risk into the green area, but the risk management team should also reduce the risk continuously.

In cases, the risk is in a land above the acceptable, but for which is possible to coexist with, related to the benefits associated and the impracticability of reducing it, it can be considered a tolerable risk.

The risk assessment will be used for both initial risk assessment and residual risk assessment.

7. Product characteristics survey questionnaire

Identify characteristics of the product that could affect Safety according to ISO 14971:2019.

QUESTIONS	ANSWERS
1. What is the intended use of the health product, and how should it be used?	VR-CARDIO is a software system intended for use by electrophysiologists and cardiology specialists outside of operating room settings. It supports the visualization and interpretation of three-dimensional (3D) electro-anatomical maps of cardiac electrophysiology, providing an immersive, three-dimensional stereoscopic view of the heart's electrical activity. This enhances non-invasive mapping capabilities during the evaluation of conditions such as Atrial Fibrillation and other arrhythmias. The software facilitates the analysis, display, and interpretation of cardiac electrophysiological data, aiding in the understanding of complex arrhythmias and guiding preliminary assessments. It is designed to assist clinicians in identifying arrhythmia mechanisms, supporting diagnosis

	and planning non-invasive management strategies. VR-CARDIO is intended to complement traditional diagnostic methods, offering advanced visualization tools to deepen clinical insights into arrhythmia mechanisms and improve the effectiveness of non-invasive cardiac care.
2. Is the health product intended to be implanted?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
3. Is the medical device intended to come into contact with the patient or other people?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
4. What materials and/or components are incorporated into the health product or used in conjunction with or in contact with it?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
5. Is energy supplied to and/or extracted from the patient?	No, VR-CARDIO is a Medical Device Software device and does not physically interact with the patient to supply or extract energy.
6. Are substances supplied to and/or extracted from the patient?	No, as a software device, VR-CARDIO does not involve the supply or extraction of any substances from the patient.
7. Is the medical device supplied sterile or intended to be sterilized by the user, or are other biological controls applicable?	No, VR-CARDIO is a Medical Device Software product and does not require sterilization or biological controls.
8. Is the medical device intended to be routinely cleaned and disinfected by the user?	No, as a Medical Device Software, VR-CARDIO does not require cleaning or disinfection.
9. Is the health product intended to modify the patient's environment?	No, VR-CARDIO operates as a software application and does not modify the patient’s environment.
10. Are there measurements to be taken?	No, VR-CARDIO processes and displays data but does not directly take measurements; it relies on data inputs from external systems.
11. Is the health product interpretive?	It is not interpretive by itself, needs the help of a cardiac signal specialist or

	electrophysiologist to make the interpretation and final conclusion.
12. Is the health product intended to be used in conjunction with drugs or other medical technologies?	No. VR-CARDIO is not intended for use in conjunction with other medicines or other medical technologies.
13. Are there unwanted outputs of energy or substances?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
14. Is the health product susceptible to environmental influences?	No. VR-CARDIO is not susceptible to environmental influences.
15. Does the health product influence the environment?	No. VR-CARDIO does not influence the environment.
16. Are there any essential accessories or consumables associated with the health product?	No. VR-CARDIO has no essential accessories or consumables associated with it.
17. Is maintenance and/or calibration required?	No. VR-CARDIO does not require maintenance or calibration.
18. Does the health product contain software?	Yes. The VR-CARDIO family is composed by 2 references of Medical Device Software.
19. Does the health product have a restricted shelf life?	No. VR-CARDIO does not have a restricted shelf life.
20. Are there any lingering and/or long-term use effects?	No. VR-CARDIO has no persistent or long-term effects.
21. What mechanical forces will the health product be subject to?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
22. What determines the useful life of medical equipment?	For VR-CARDIO, as a software-only product, the useful life is determined by the software's ability to remain functional, secure, and compliant with evolving regulatory and technological standards. This includes factors such as ongoing software updates, compatibility with operating systems, and continued support for cybersecurity measures. Regular maintenance, user feedback, and market surveillance ensure the software's performance and usability over time.

23. Is the health product intended for single use?	No. VR-CARDIO is intended for multiple uses
24. Is it necessary to take special safety measures or deactivate the health product?	No. VR-CARDIO is a Medical Device Software and does not have physical components.
25. Does installation or use of the medical device require special training or special skills?	Yes, training is provided by the SPIKA TECH S.L. team. VR-CARDIO Explore is primarily a web application and does not require installation in that format. However, for VR-CARDIO VHET360, installation is required. In such cases, clear and detailed installation instructions are provided to the user to ensure a smooth setup process.
26. How will security information be provided?	By messages on the screen and are available in the Instructions for Use and Labelling.
27. Will there be a need to establish or introduce new production processes?	No, at the moment it has not been necessary.
28. Is the successful application of the health product critically dependent on human factors such as the user interface?	Yes, it is dependent on the human-machine interface.
28.1 Can user interface design features contribute to usage errors?	Yes, those errors are residual risks that are included in the IFU.
28.2 Will the health product be used in an environment where distractions can cause usage errors?	No, it is used in specific environments for this purpose.
28.3 Does the health product have parts or accessories connected to it?	No, as a software-only product, VR-CARDIO does not inherently include or rely on any physical parts or accessories for its primary functionality. It operates independently as a web-based application or desktop software. However, optional accessories (see B2.8 Accessories) may be connected to enhance the user experience, such as providing additional visualisation capabilities.
28.4 Does the medical device have a control interface?	It`s controlled through the workstation device (computer) peripherals.

28.5 Does the health product display information?	Yes, it provides information to the healthcare professional validates or rejects the information provided.
28.6 Is the health product controlled by a menu?	Yes, there is a navigation menu.
28.7 Will the health product be used by people with special needs?	Not applicable. It is intended to be used by VR-CARDIO.
28.8 Can the user interface be used to initiate user actions?	No. Any restricted mode is password protected.
29. Does the health product use an alarm system?	No, VR-CARDIO does not use an alarm system.
30. In what way(s) can the health product be deliberately misused?	Used without proper training and by people without professional knowledge.
31. Does the health product store data critical to patient care?	Yes, they are stored securely for later analysis and interpretation by a healthcare professional in the area.
32. Is the health product designed to be mobile or portable?	No, VR-CARDIO as a software-only product is not inherently mobile or portable. However, it can be accessed from compatible devices such as desktop computers or laptops, allowing flexibility in its use across different locations. Its portability is dependent on the hardware used to access the software.
33. Does the use of the medical product depend on essential performance?	Yes, depends on the algorithm's performance.

8. Analysis, assessment and risk management

Risk management is an essential stage in the life cycle of a medical device. This analysis highlights the potential hazardous phenomena associated with the device, as well as identifying the underlying causes.

This assessment determines whether, or not, the risks identified are considered acceptable. Risk acceptability criteria may vary depending on the specific medical device and the applicable regulations.

The next phase consists of implementing risk control measures. Appropriate actions are taken to reduce the identified risks to their lowest possible level. These measures may

include design improvements, the addition of protective devices or the provision of clear warnings to users.

Once these measures have been implemented, the residual risk is assessed. This is the level of risk that remains once the risk reduction measures have been applied. This assessment is followed by a benefit/risk analysis, which examines whether the potential benefits of the medical device outweigh the residual risks identified.

Finally, an overall assessment of the acceptability of all residual risks is carried out. This assessment considers the individual residual risks as well as their overall contribution to the safety and performance of the medical device. It determines whether the level of residual risk is considered acceptable in the specific clinical context.

9. Risk analysis

Risk analysis was performed as described according to ISO 14971, clauses 5.1 to 5.5.

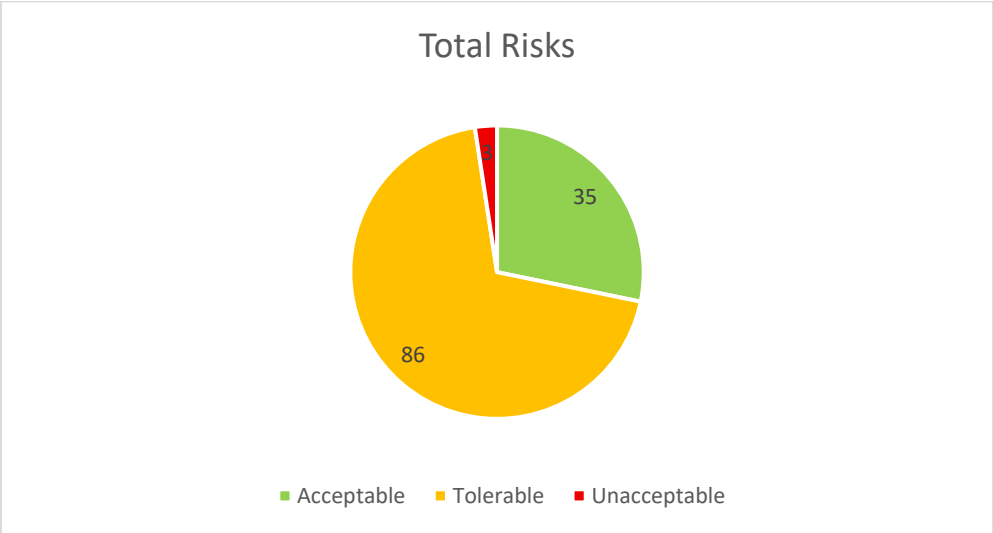
The intended use and identification of characteristics related to the safety of the medical device are identified based on the questions from Annex A, ISO/TR 24971:2020.

The potential hazards identified are usually related to usability, data/record maintenance, design and development, cybersecurity and SOUPs.

Before applying and implementing Risk Control Measures, the total count of risks in the Risk Matrixes (see Annex I and Annex II):

- 35 acceptable risks
- 86 tolerable risks with measures
- 3 unacceptable risks

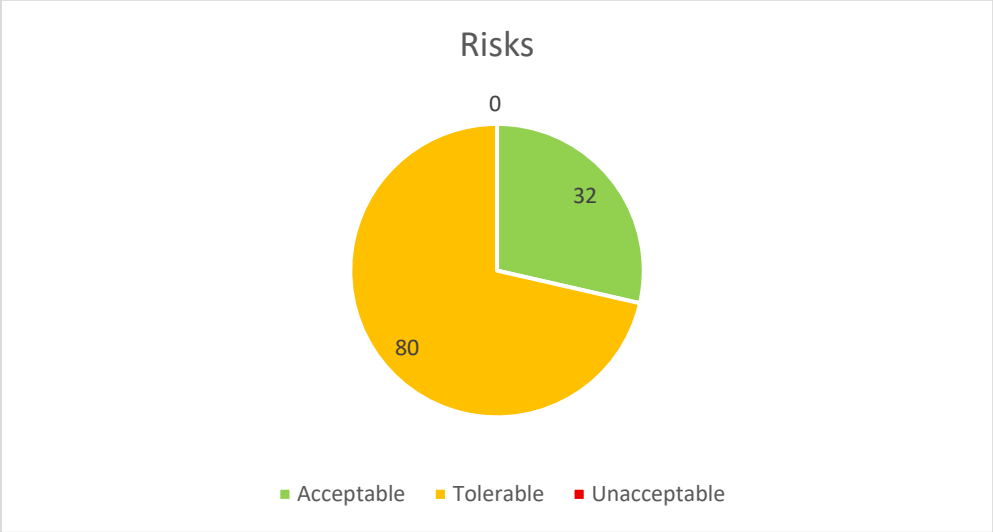
Total of 124 risks analysed



Taking into account the risks described in Annex I, the count of risks in the Risk Analysis Matrix was as follows:

- 32 acceptable risks
- 80 tolerable risks, acceptable risks with measures
- 0 unacceptable risks

Total of 112 risks analysed



Regarding SOUPs specific risks (see Annex II), the count of risks in the Risk Matrix was as follows:

- 3 acceptable risks

- 6 tolerable risks, acceptable risks with measures
- 3 unacceptable risks

Total of 12 risks analysed

10. Risk control measures

Risks were reduced as far as possible (AFAP). If a risk was classified as “tolerable” or “Unacceptable” based on the Risk Analysis, Risk Control Measures shall be implemented. Potential categories of Risk Control Measures are “inherent safety by design”, “protective measures”, and “information for safety”.

A total of 124 risk control measures were implemented, with all 124 risks being reduced to AFAP.

Taking into account the risks described in the Annex I, a total of 112 risk control measures were implemented with all 112 risks being reduced to AFAP. The control measures include reference to the Instructions for Use, Training, etc.

Regarding SOUPs specific risks (see Annex II), 12 risk control measures were implemented for all the risks, with all 12 risks being reduced to AFAP. The control measures include reference to the System, Unit, Integration and Performance tests.

11. Risk acceptability

After implementation and verification of all Risk Control Measures, the count of risks in the Risk Acceptance Matrices was as follows:

- 124 acceptable risks
- 0 Tolerable
- 0 unacceptable risks

Total of 124 risks

Of the measures taken to reduce the risk, 86 of the risks identified as “tolerable”, that is, acceptable but with additional measures, 86 were reduced to an acceptable level. A total of 6 risks identified as “Unacceptable” were also reduced to an acceptable level. Furthermore, 35 of the risks identified as “acceptable” were reduced AFAP.

12. Risk resulting from risk control measures

No new risks were created after the implementation of risk control measures. The implementation of the risk management plan has led to the elimination of all risks considered “unacceptable”. Appropriate control measures have been put in place to minimize all risks to the lowest possible level.

After the implementation of control measures the risks have been reduced or eliminated to the minimum possible by the manufacturer, and the manufacturer informs the user of residual risks associated with the manipulation of the IFU.

13. Summary of risk and unacceptable risks

After Risk Control Measures, all existing risks were considered acceptable. The device therefore, fulfils the specifications of the defined risk policy and is safe.

14. Benefit/risk assessment

Benefits	VR-CARDIO family: • Provides detailed and accurate visualization of the heart's electrical activity, supporting accurate diagnosis and more effective treatment planning. • Offers an immersive, three-dimensional stereoscopic view and 360° holographic visualization of the heart's electrical activity. • Enhances non-invasive mapping capabilities for conditions like Atrial Fibrillation and other arrhythmias, reducing the need for invasive procedures and associated risks like radiation exposure. • Aids in identifying arrhythmia mechanisms and supporting diagnosis and non-invasive management strategies. • Complements traditional diagnostic methods and overcomes limitations of two-dimensional systems, enabling deeper analysis and understanding of heart structure and function. • Enables near real-time monitoring of the heart's electrical activity, improving the efficiency of cardiac care procedures.
Risks	• The primary risks identified are related to data quality and user interpretation, such as inaccurate results from incorrect patient registration or data acquisition, data loss due to internet connection issues, or performance loss from inadequate hardware. • There is a risk of over-reliance on the algorithm's predictions, which is mitigated by the explicit requirement for medical professional validation of all diagnostic information and decisions. • Cybersecurity risks related to data privacy and unauthorized access are addressed through login methods, session tokens, and data encryption.

	• The device's classification as Class IIa implies that decisions based on its output should not cause death or an irreversible/serious deterioration of a person's state of health, nor require surgical intervention. The software poses no immediate danger to patients.

15. Overall residual risk

All situations with any danger and potential damage identified were evaluated and the risks were reduced to its lowest possible level. The overall residual risk associated with the medical device meets the global residual risk acceptability criteria as there are no individual residual risks.

All risks were considered acceptable and additional control measures are not practicable.

After comparing the medical benefits obtained from the use of the device with the risks, the evidence collected allows us to conclude that the expected benefits clearly outweigh the risks and, therefore, the device constitutes added value and should be made available on the market.

Use of the product is free from unacceptable risks when using the device in accordance with the manufacturer's instructions for use. The medical device does not raise undue safety concerns; however, the safety and performance of the device will be continually monitored through ongoing surveillance activities.

16. Conclusion

Based on the established risk management plan, having carried out an in-depth analysis of the risks associated with VR-CARDIO family and implementing appropriate control actions, we can conclude that the risks are now under control. The risk management measures implemented have effectively reduced the potential risks associated with the use of the VR-CARDIO family.

It is essential to highlight that patient safety and compliance with quality standards were taken into consideration in the management of these risks.

However, it is important to maintain ongoing monitoring and carry out regular assessments to ensure that risk control measures remain effective over time. This proactive approach will help ensure device safety and quality during clinical practice.

17. Revisions history

Version	Change description	Prepared by	Approved by	Date
01	First edition.	Iván Pérez	Fernando Pozo	30/11/2024
02	Document name change and information update	Iván Pérez	Fernando Pozo	11/06/2025

18. Annexes

Annex I – Risk Analysis

Annex II – SOUP List